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W3 Data Visualisation

Lecture

Class

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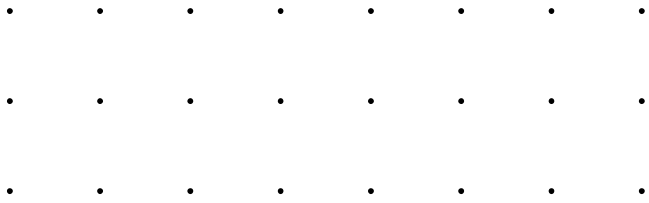
Agenda

- 3.1 Storytelling
- 3.2 Data governance and privacy

Agenda

- Audience and purpose
- Storyboarding
- Creating charts
- Design critique
- **DEMONSTRATION 1**

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W3.1 COS30045

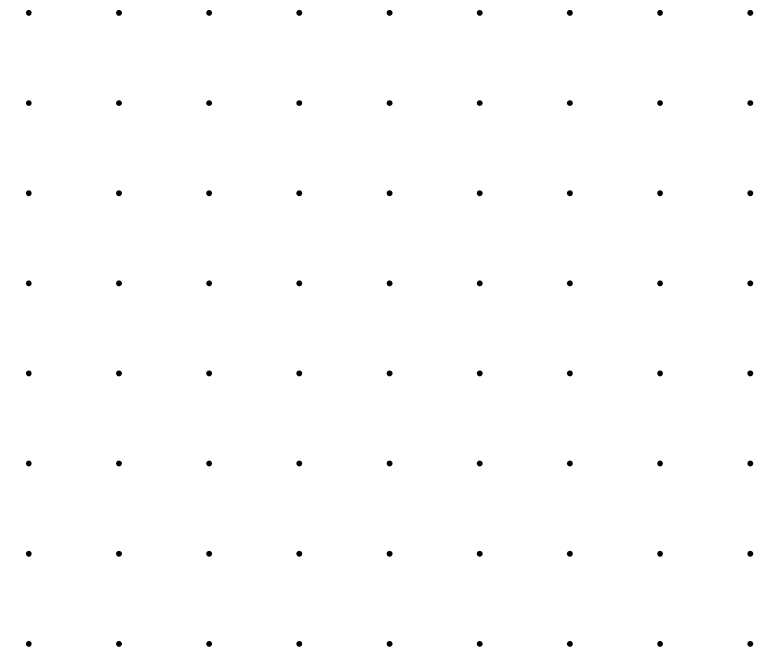
Lecture

Agenda

- Storytelling

Reference

- Knaflitz (2015) *Storytelling with data*
- Ch 1&2



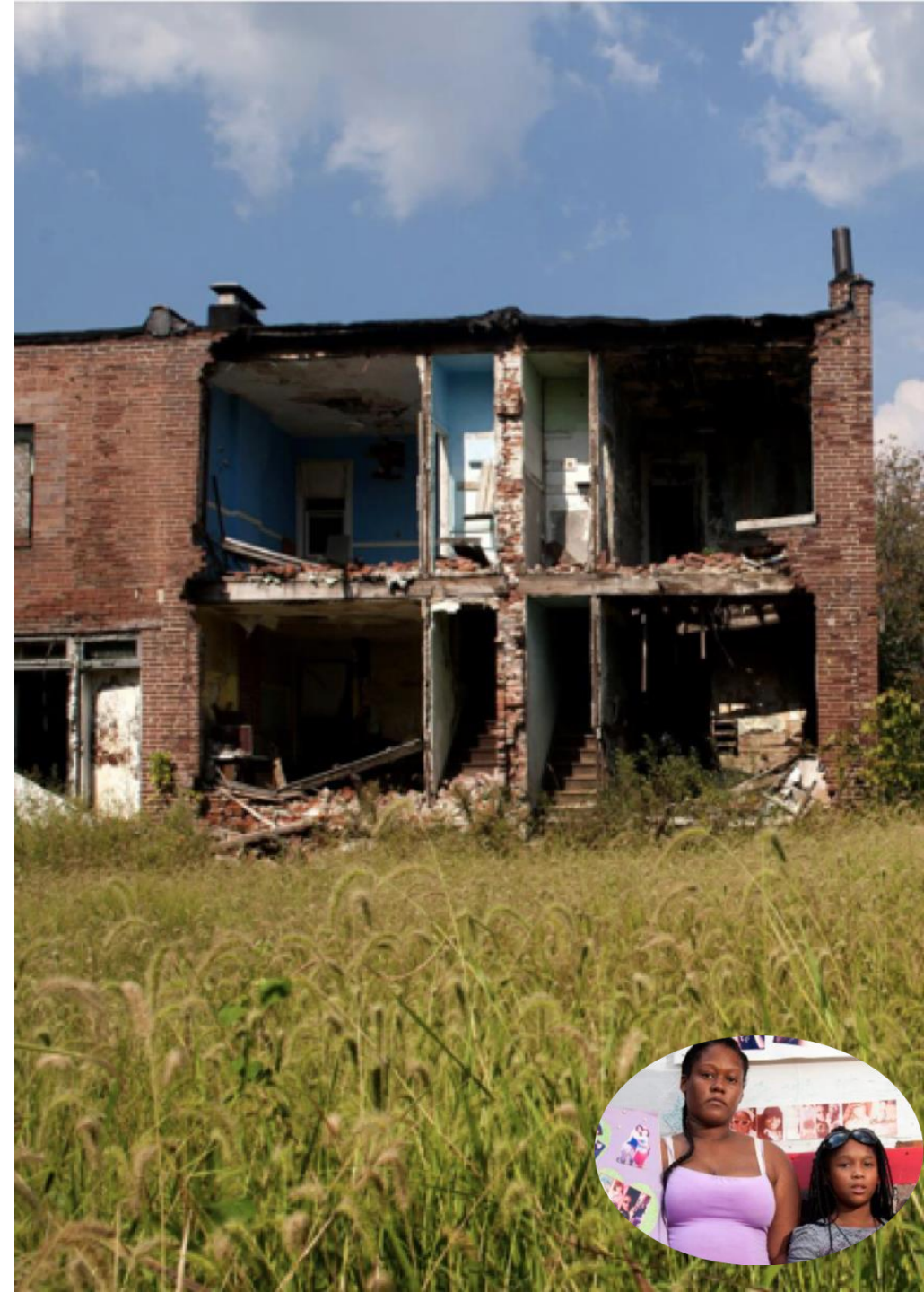
Exploratory vs Explanatory

Exploratory:

- finding the patterns, working out what the data has to say

Explanatory:

- communicating what we found in the data, what were the interesting things



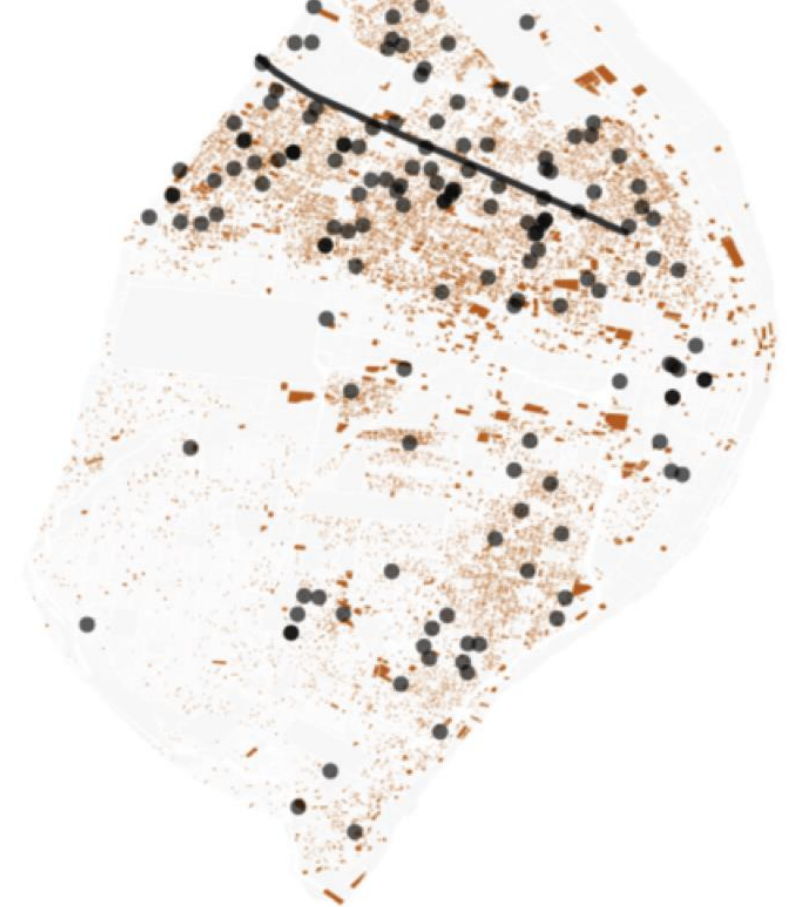
Exploratory vs Explanatory

Exploratory:

- finding the patterns, working out what the data has to say

Explanatory:

- communicating what we found in the data, what were the interesting things



St Louis

■ Condemned parcel, excluding parks

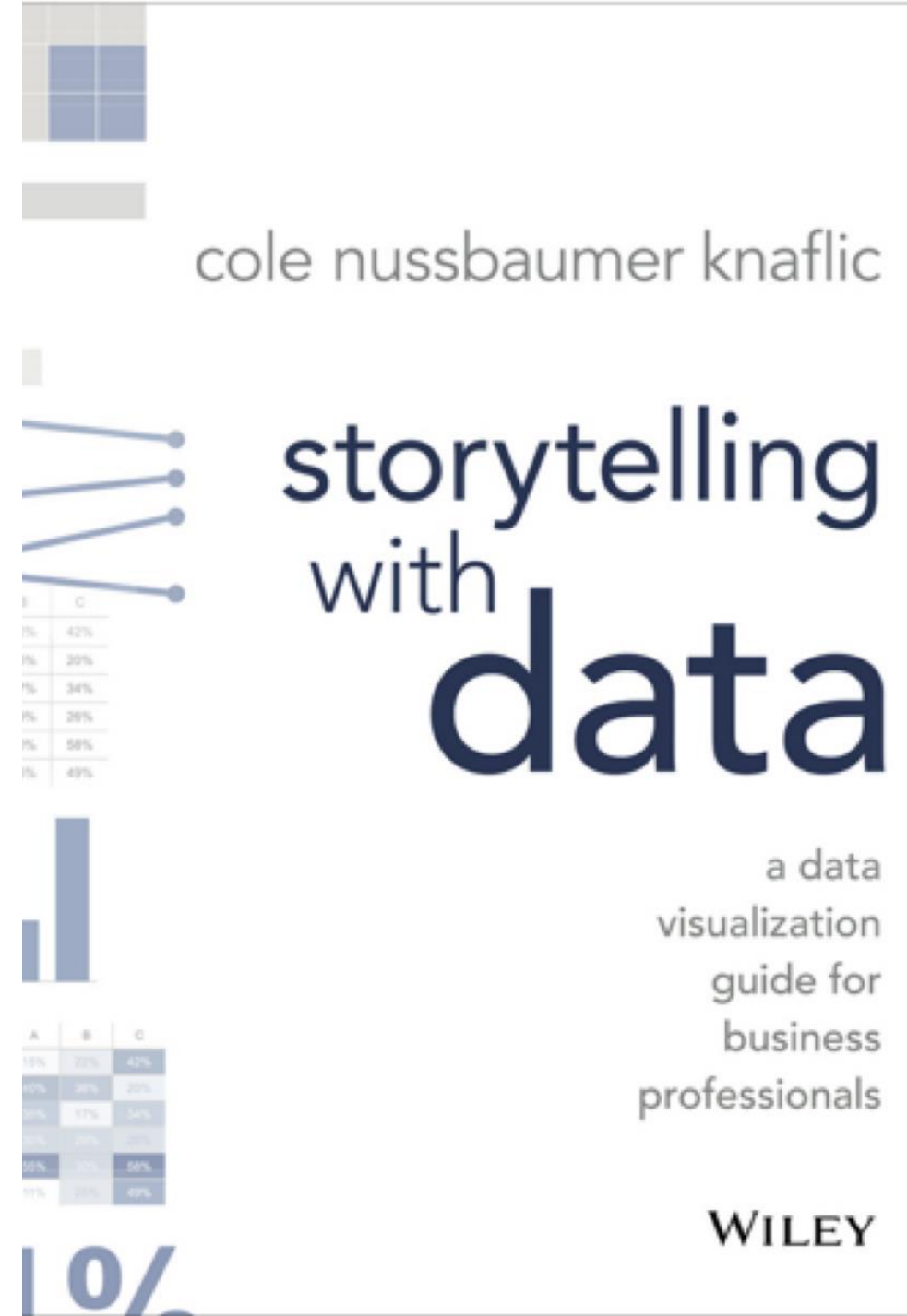
● 2015 gun homicide incident

— Natural Bridge Rd.

Storytelling with Data

Knaflic (2015) data vis with business bent

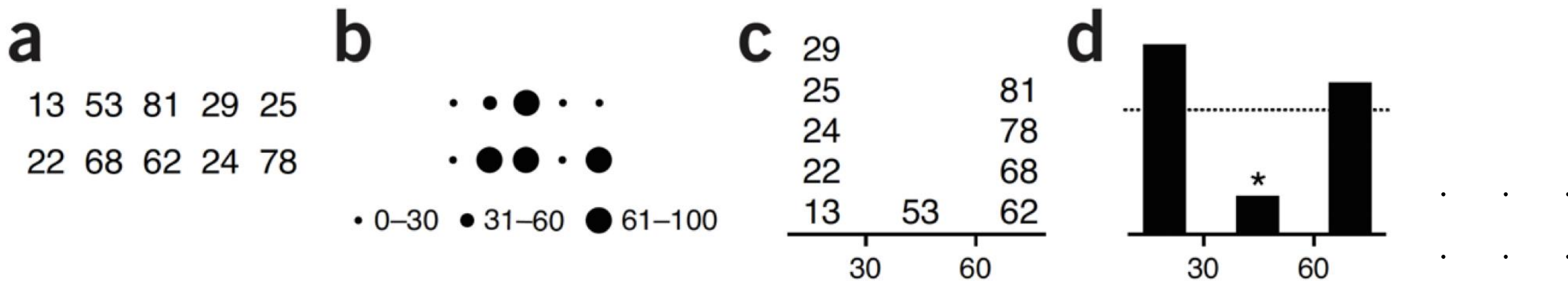
- focus on informing and convincing audience to some action (not 'scientific'/objective/let the audience decide presentation)
- clear and effective communication
- fine line between 'showing the important data' vs obscuring or manipulating presentation to fit the message



Role of Story Telling in Science

“Inviting readers to draw their own conclusions is risky because even simple messages can hide in simple data sets”

“Maintain focus of your presentation by leaving out detail that does not advance the plot”



Role of Story Telling in Science

“Storytelling encourages the unrealistic view that scientific projects fit a singular narrative.”

“Storytelling... impedes communication of critical information by restricting the scope of the data to that agreeable with the story.”



Katz (2013)

“Maintain focus of your presentation by leaving out detail that does not advance the plot”



Krzywinski & Cairo (2013)

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Who, What and How

Who is the audience?

- job role (marketing, business manager, product dev...)

What is your relationship to audience?

- part of on-going team
 - less need to explain method
- outside consultant/agency
 - more need to explain method
 - establish credibility



What do you want the audience to know?

- Don't leave it up to the audience to interpret the data
- Help the audience find the message in the data
- Get audience to take some action

accept | agree | begin | believe |
change | collaborate |
commence | create | defend |
desire | differentiate | do |
empathize | empower | encourage
| engage | establish | examine |
facilitate | familiarize | form |
implement | include | influence |
invest | invigorate | know | learn |
like | persuade | plan | promote
| pursue | recommend | receive |
remember | report | respond |
secure | support | simplify | start
| try | understand | validate

Knaflic(2015)

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Planning your message

3 minute Story (aka Elevator Pitch)

- if you only had 3 min to tell the audience what they need to know...

The Big Idea (Duarte cited Knaflitz)

- “It must articulate your unique point of view;
- It must convey what’s at stake; and
- It must be a complete sentence.”



Story Telling - The basics

In the beginning (setting)...

- introduction to the main character (audience) and setting

The conflict...

- an incident occurs that changes something of importance for the character (imbalance)
- the character has to resolve the problem (re-balance)
- learn new skills, deal with situation, make change, ...

The resolution...

- call to action - what the audience needs to do
- balance is restored

<https://www.scifiworld.es/imagenes/star-wars-a-new-hope-1977.html>



Storyboard

Plan story with post it note rather than presentation software



FIGURE 1.2 Example storyboard

Decide how to present the data

Pilot program was a success

After the pilot program,

68%

of kids expressed interest towards science,
compared to 44% going into the program.

Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.29 Show the numbers directly

Decide how to present the data

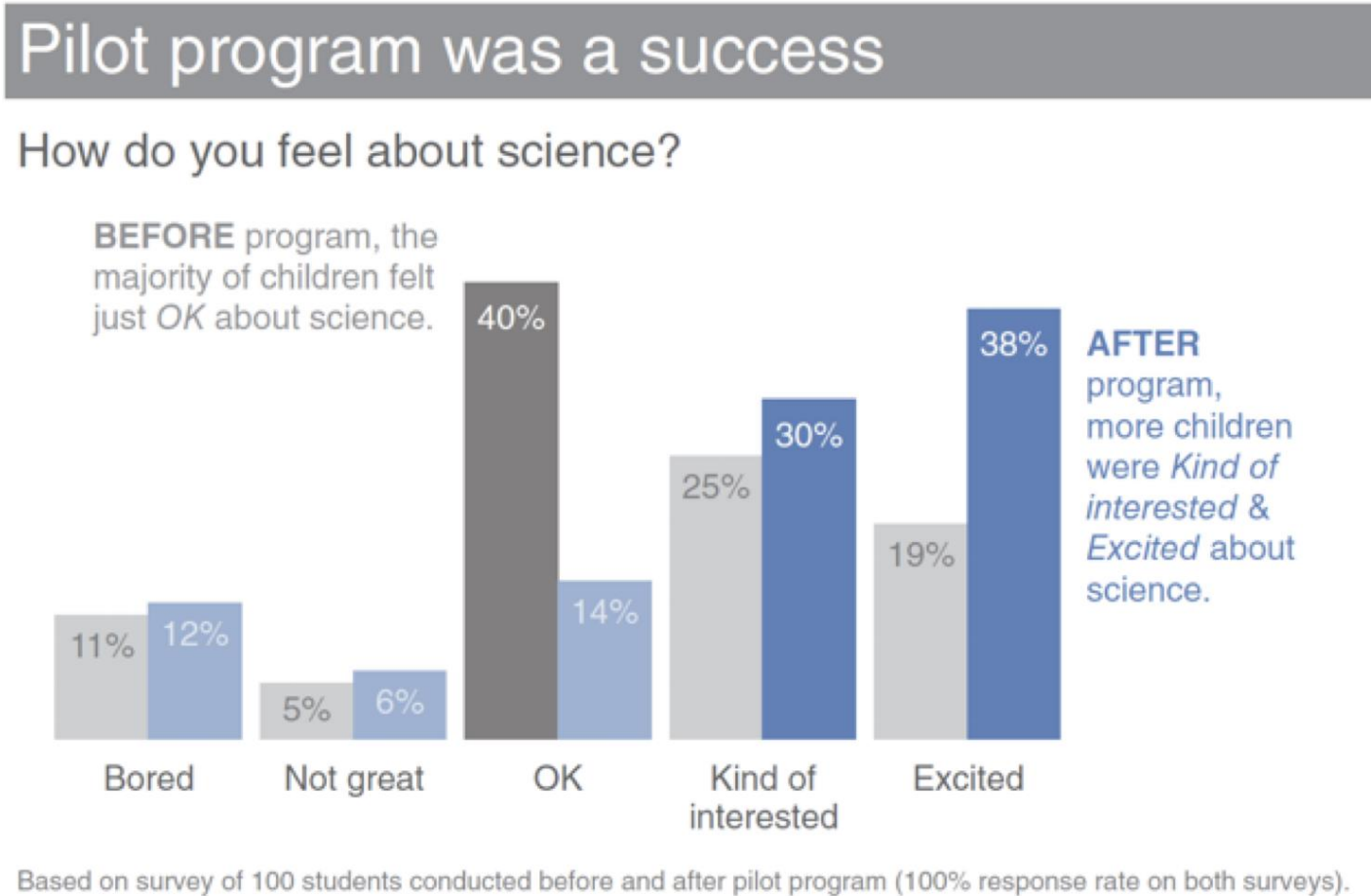
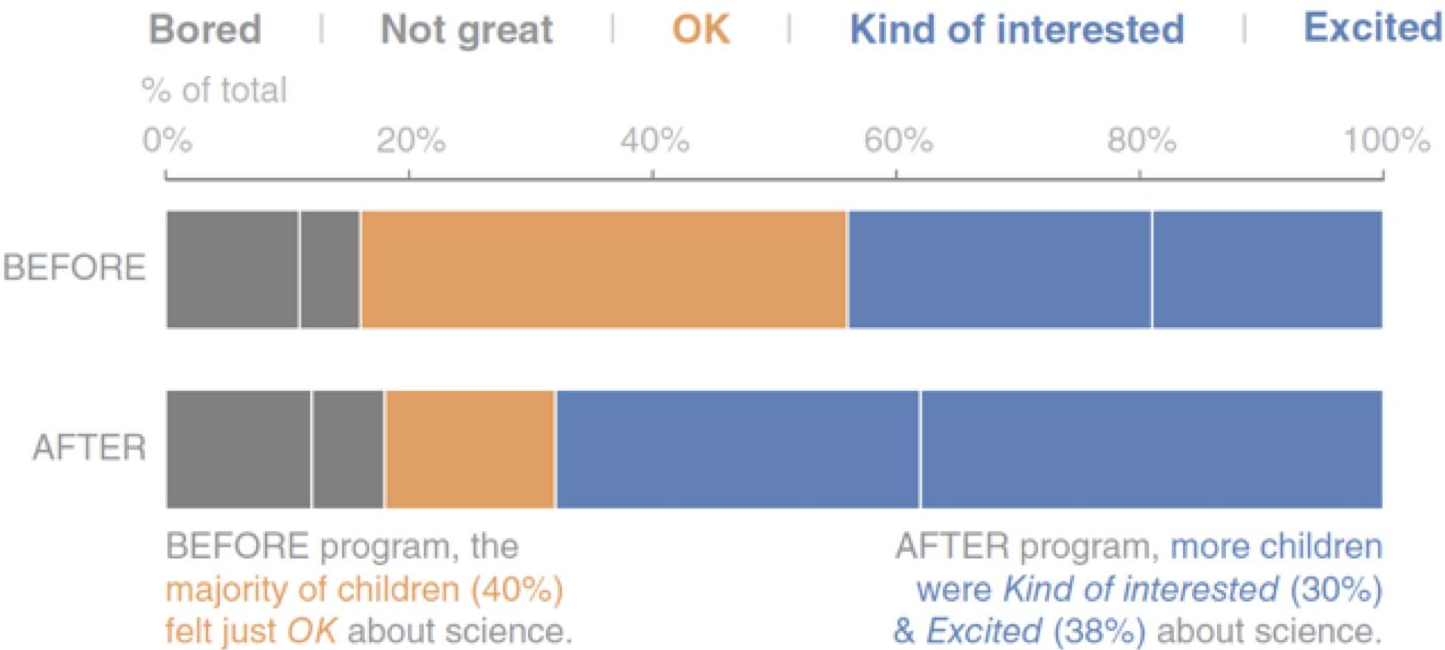


FIGURE 9.30 Simple bar graph

Decide how to present the data

Pilot program was a success

How do you feel about science?



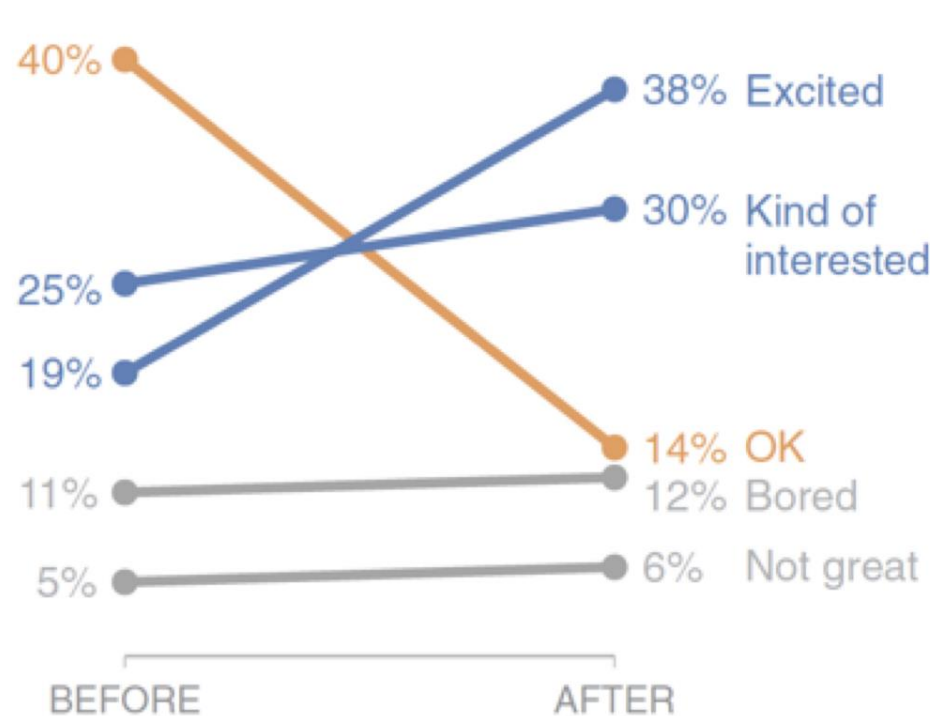
Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.31 100% stacked horizontal bar graph

Decide how to present the data

Pilot program was a success

How do you feel about science?



BEFORE program, the majority of children felt just *OK* about science.

AFTER program, more children were *Kind of interested & Excited* about science.

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Based on survey of 100 students conducted before and after pilot program (100% response rate on both surveys).

FIGURE 9.32 Slopegraph

Decide how to present the data

Peak Break-up Times

According to Facebook status updates

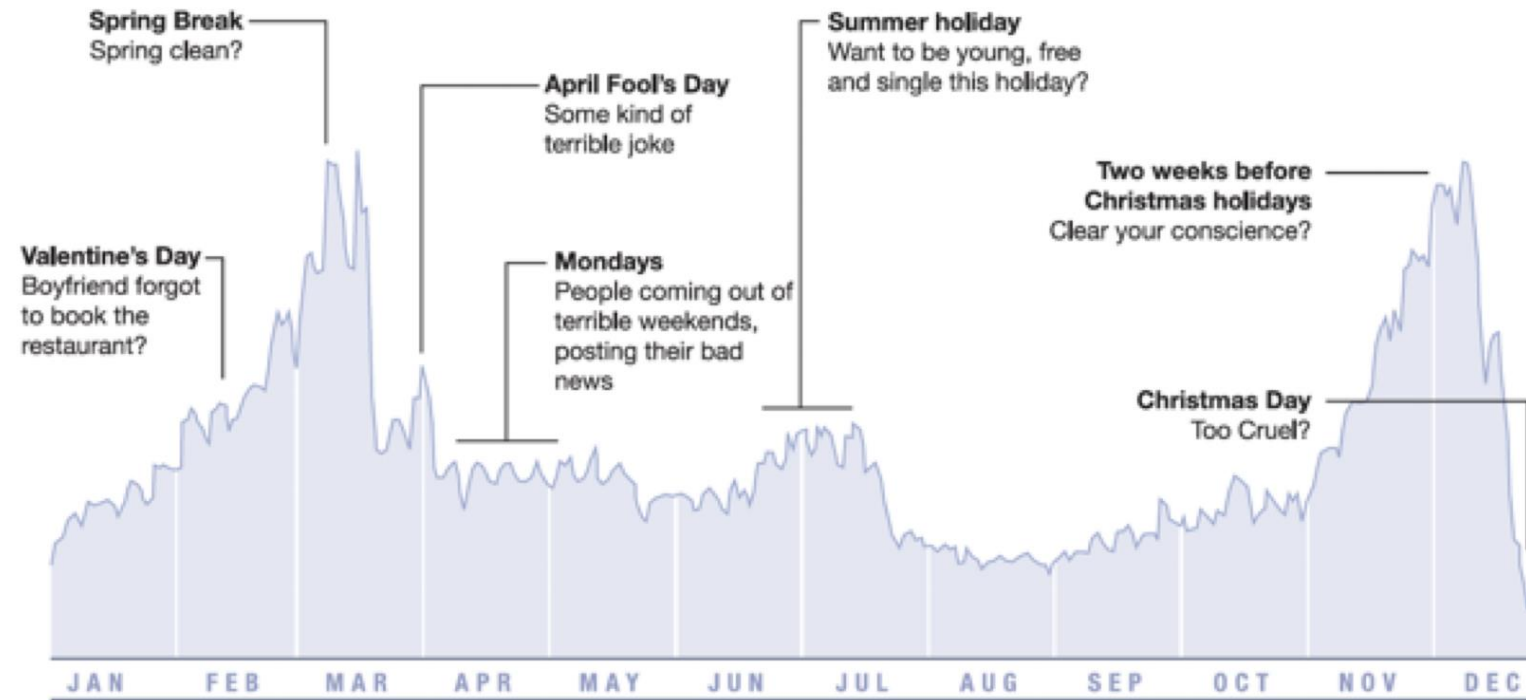


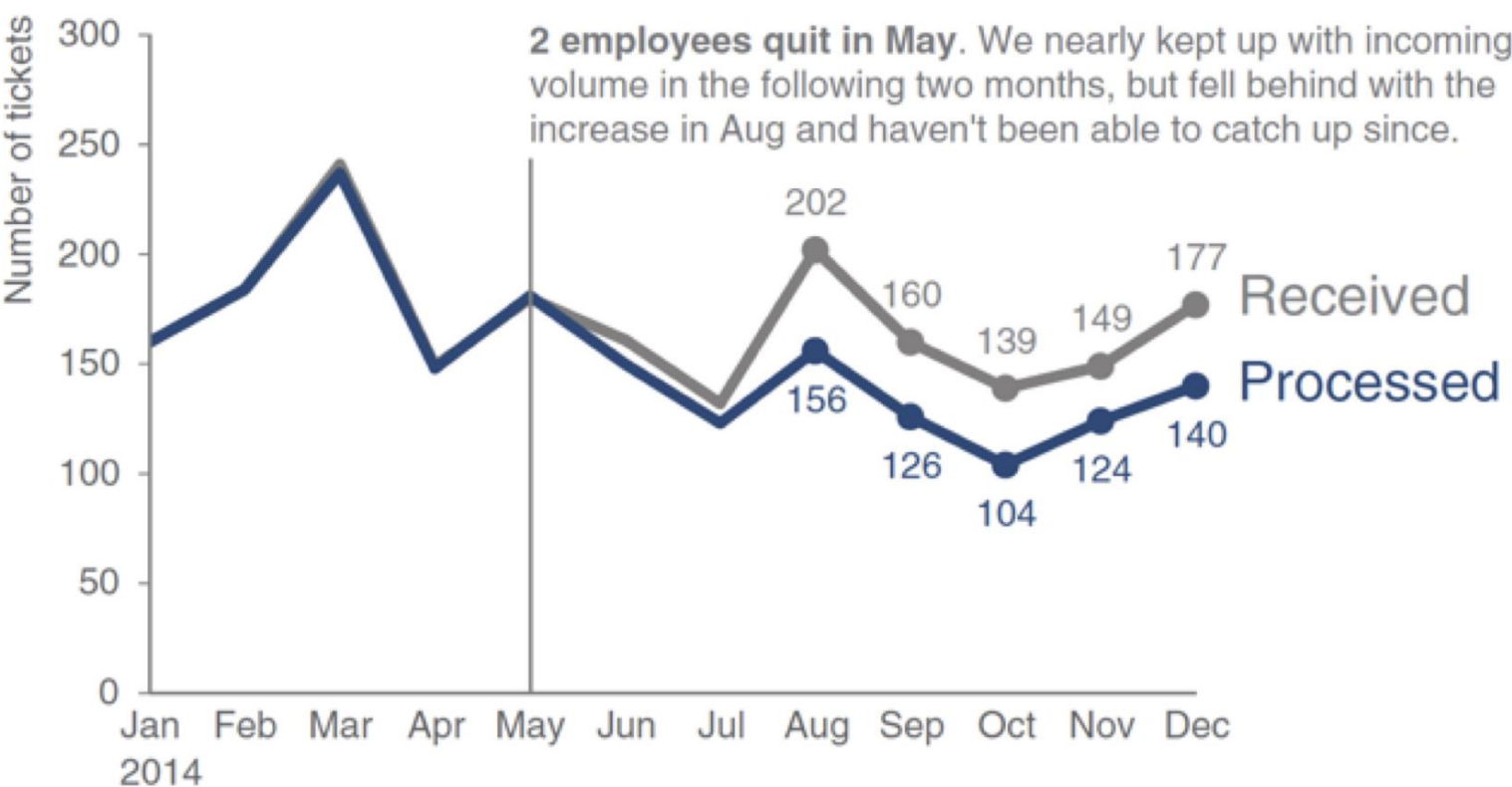
FIGURE 5.7 Words used wisely

Decide how to present the data

Please approve the hire of 2 FTEs

to backfill those who quit in the past year

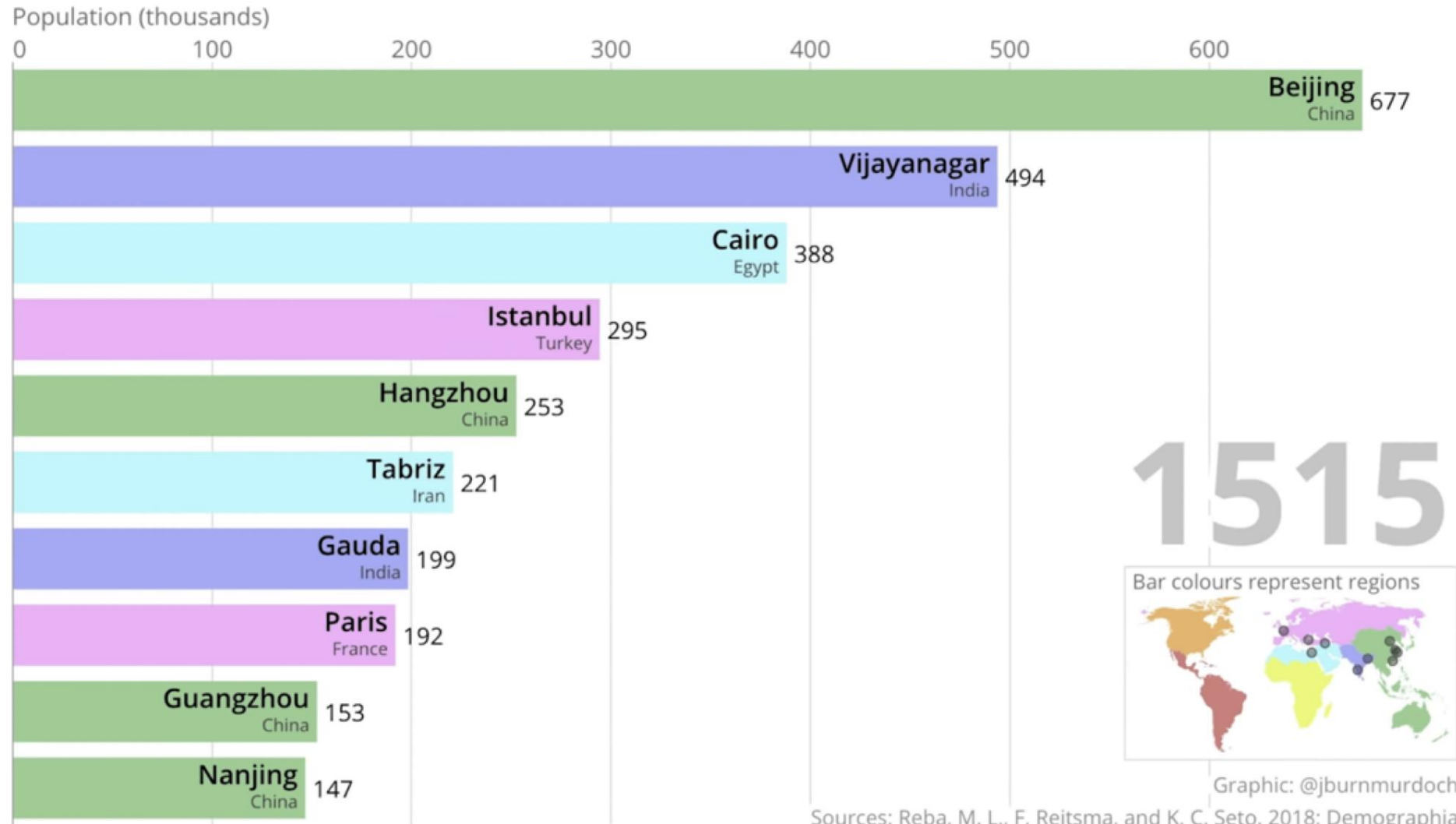
Ticket volume over time



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The most populous cities in the world from 1500 to 2018



I would have written a shorter letter, but I did not have the time.



Blaise Pascal (1656)

French Mathematician, logician,
physicist and theologian

